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A resource orchestration perspective on longitudinal product innovation studies: a systematic review and future research framework

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Abstract

Purpose – This study aims to review and synthesize longitudinal research on product innovation management and to develop a research agenda using a resource orchestration perspective. Given the absence of comprehensive reviews over the past decade, the study seeks to advance understanding of how innovation processes unfold over time and to identify underexplored areas in the literature, particularly regarding the role of resource structuring in enabling sustained product innovation.

Design/methodology/approach – A systematic literature review was conducted resulting in the inclusion of 219 longitudinal studies on product innovation management. Studies were selected based on predefined inclusion criteria, including a minimum time span of three years. The analysis combined methodological classification with a theory-driven coding approach based on the resource orchestration framework, examining how structuring, bundling and leveraging processes have been addressed in both qualitative and quantitative research.

Findings – The findings show that most longitudinal studies focus on aggregated innovation outcomes and emphasize resource bundling and leveraging, while resource structuring remains underexamined. In particular, the configuration and development of human resources are largely overlooked despite their importance for long-term innovation. The review also reveals limited integration of orchestration processes over time, highlighting a lack of holistic perspectives on how structuring, bundling and leveraging interact dynamically in innovation processes.

Originality/value – This study provides an updated systematic review of longitudinal product innovation management research and offers a resource orchestration-based research agenda. By identifying resource structuring – especially human resource configuration – as a critical but neglected area, the study extends current innovation research and offers a detailed, theory-driven agenda for future longitudinal studies.

Keywords Longitudinal study, Literature review, Resource-based theory, Product innovation, Resource orchestration, Sustaining Innovation, Systematic Innovation

Paper type Literature review

1. Introduction

The management of product innovation is a long-term endeavor and there are several reasons to use longitudinal approaches in innovation studies (Perks and Roberts, 2013). Specifically,

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product innovation is mostly regarded as a process (Vuori and Huy, 2016) and managing this process generally requires the long-term synchronization of various activities (Carnes *et al.*, 2022). Moreover, innovation management concerns the overall management of innovation rather than isolated innovation processes, requiring an understanding of how innovation activities unfold and interact over time. In addition, there is generally a time between the launch of a new product innovation and its impact on firm performance (Frenz and Ietto-Gillies, 2009), necessitating a longitudinal approach to address the result of innovation management practices.

However, relatively little attention has been devoted to synthesizing longitudinal product innovation studies from a resource management perspective. The latest review explicitly focusing on longitudinal product innovation studies was conducted over a decade ago (Perks and Roberts, 2013). Since then, innovation research has expanded considerably, including reviews addressing themes such as dynamic capabilities (Laaksonen and Peltoniemi, 2018; Alwali, 2024), absorptive capacity (Zou *et al.*, 2018) and organizational ambidexterity (Tarba *et al.*, 2020). However, these reviews have generally focused on specific theoretical perspectives or concepts rather than examining how resources are managed throughout longitudinal product innovation processes. Resource orchestration (Sirmon *et al.*, 2007; Sirmon *et al.*, 2011) has attracted increasing scholarly attention in innovation management studies (Carnes *et al.*, 2022; Carnes *et al.*, 2017; Chirico *et al.*, 2024; Andersén and Ljungkvist, 2021). It provides a particularly suitable analytical lens for synthesizing longitudinal product innovation studies by encompassing resource management processes from the development of resource portfolios to the deployment of capabilities for innovation. These interconnected processes unfold over different temporal horizons, ranging from long-term resource acquisition, accumulation and divestment to the bundling, mobilization, coordination and deployment of capabilities for product innovation. Consequently, resource orchestration provides a comprehensive analytical framework for identifying which resource management processes have received scholarly attention and, equally importantly, which remain underexplored. Moreover, resource orchestration builds on some of the seminal traditions in innovation research, tracing back to the recombination of resources described by Schumpeter (1934) and the evolutionary approach developed by Nelson and Winter (1982). Accordingly, there is a need for an updated synthesis of longitudinal product innovation studies that applies a resource orchestration perspective to identify neglected resource management processes and guide future longitudinal innovation research.

The purpose of this article is to present an overview of longitudinal studies on product innovation management and to, based on a resource orchestration framework, propose avenues for future longitudinal product innovation management research. Based on a review of 219 studies, this study makes two main contributions. First, it provides an updated synthesis of longitudinal product innovation studies published across innovation and management journals. Second, by interpreting this body of research through a resource orchestration perspective, it identifies neglected orchestration processes and develops a theory-informed agenda for future longitudinal innovation research.

2. Theoretical framework

2.1 Defining longitudinal product innovation management

We define product innovation as novelty in terms of new products or renewal of existing products and that "the ultimate source of novelty in successful innovation lies in the recombination of conceptual and physical resources" (Carnes *et al.*, 2022: 162). As argued by Perks and Roberts (2013), innovation is generally regarded as a process of several events and examining an isolated episodic event is not enough to fully grasp the complexity of

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innovation processes (Sørensen *et al.*, 2010). Studies addressing organizational-level outcomes in research domains related to innovation generally define longitudinal studies as studies examining several years. For example, in their study on resource immobility, Andersén *et al.* (2016) defined a longitudinal study as a study covering a time period of five years or more. Thus, we argue that a longitudinal study on product innovation management, especially when analyzed from a resource orchestration approach, should encompass several years and we therefore adopt three years as an inclusive lower boundary for longitudinal research rather than as a theoretically optimal duration. Although longer observation periods are generally preferable for capturing cumulative resource orchestration processes, we adopted a minimum threshold of three years to avoid excluding studies that examine meaningful longitudinal innovation processes while maintaining a clear distinction from short-term or cross-sectional research.

2.2 Resource orchestration and innovation management

Viewing innovation as the recombination of resources is one of the most established perspectives in innovation management research (Schumpeter, 1934; Carnes *et al.*, 2017). Accordingly, resource orchestration has attracted considerable attention as a framework for analyzing resource management processes in innovation (e.g. Andersén, 2021; Carnes *et al.*, 2022; Carnes *et al.*, 2017). Resource orchestration "is the comprehensive process of structuring the firm's resource portfolio, bundling the resources to build capabilities, and leveraging those capabilities with the purpose of creating and maintaining value" (Sirmon *et al.*, 2007: 273) [we have italicized the three main processes of the framework]. Structuring is about the *acquisition* of resources in factor markets, the *accumulation* of resources within organizations and the *divestment* of redundant resources. Bundling concerns the development of new organizational capabilities through pioneering, the enrichment of existing capabilities and the stabilization of current capabilities. In the leveraging process, capabilities are first *mobilized* by identifying configurations of capabilities necessary to exploit market opportunities. These configurations are then *coordinated* with other capabilities and resources and lastly physically *deployed* to generate value.

While related perspectives such as absorptive capacity (Cohen and Levinthal, 1990; Zahra and George, 2002; Mady *et al.*, 2022; Al-Husseini, 2024), dynamic capabilities (Teece *et al.*, 1997; Hutton *et al.*, 2021; Ferreira and Coelho, 2020) and organizational ambidexterity (Tarba *et al.*, 2020; Voss and Voss, 2013; Corchuelo Martínez-Azúa *et al.*, 2025) have generated important insights into innovation management, they primarily emphasize particular aspects of innovation, including the acquisition and utilization of external knowledge, adaptation and resource reconfiguration in response to changing environments and the strategic balance between exploration and exploitation, respectively. Resource orchestration has a different analytical emphasis by explicitly addressing how firms structure their resource portfolios, develop these resources into capabilities and leverage the resulting capabilities. Importantly, the structuring and bundling processes explicitly recognize that the accumulation and development of resources and capabilities may unfold over extended periods, rather than primarily involving their reconfiguration in response to immediate environmental changes. However, the comprehensiveness of resource orchestration also entails limitations, as the framework provides less detail regarding some specific resource management challenges. For example, although it encompasses both internal and external resources, it does not explicitly distinguish between the challenges associated with orchestrating these different types of resources (Andersén, 2023). Taken together, its explicit attention to resource structuring, capability development and capability leveraging makes

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resource orchestration particularly suitable for examining how resource management processes evolve over time and ultimately contribute to innovation outcomes.

Innovation studies have applied resource orchestration in different ways. Some innovation studies have applied specific processes from the framework to examine how orchestration influences innovation outcomes (Carnes and Ireland, 2013; Carnes *et al.*, 2017). However, most studies on resource orchestration and innovation have applied the entire orchestration framework by applying it as an aggregated construct or by examining all, or several, orchestration processes. Some of these studies have adapted the model to specific contexts, such as orchestration with suppliers to promote innovation (Chirico *et al.*, 2024) or how information technology resources can be orchestrated for innovation (Cui *et al.*, 2022). Other studies have applied more generic approaches to resource orchestration and innovation by, for example, examining how synchronization of all orchestration processes affects innovation (Carnes *et al.*, 2022). However, despite its inherently longitudinal nature, empirical applications of resource orchestration have predominantly relied on cross-sectional (e.g. Carnes *et al.*, 2022) or relatively short-term research designs (e.g. Andersén and Ljungkvist, 2021). This suggests considerable potential for applying the framework to better understand how different resource management processes interact and evolve throughout product innovation over time.

3. Methodology

3.1 Data collection

The data collection procedure is summarized in Figure 1. Following the main stages of the PRISMA procedure, the flow diagram reports the identification, screening, full-text assessment and final inclusion of studies, and shows how the application of the search and eligibility criteria resulted in the final sample.

We used Web of Science as the data source because it offers a comprehensive database of scientific publications and citations across all disciplines and it includes all journals examined in this review. The complete Web of Science search strategy, including search terms and journals, is provided in Appendix A1. Following established practices of many management reviews (Newbert, 2007; Andersén *et al.*, 2016), we limited the search to full-length peer-reviewed journal articles written in English. The search terms were applied to article titles, abstracts and keywords. We used "longitudinal" as the principal search term because the review focuses on studies explicitly designed and presented as longitudinal. Although studies employing alternative terminology may also have been relevant, using "longitudinal" as the principal search term provided a consistent and transparent basis for identifying studies explicitly examining innovation processes over time. As the review was restricted to Web of Science-indexed journals, relevant studies published in other databases or non-indexed outlets may not have been captured. However, this criterion ensured a consistent level of publication quality and transparency throughout the review. Given the purpose of identifying broader patterns in the attention devoted to different resource orchestration processes rather than exhaustively cataloguing all longitudinal innovation studies, we consider the final sample of 219 studies sufficiently broad to provide a robust basis for the synthesis.

In the first search, we included the innovation journals used in the latest review of longitudinal innovation studies (Perks and Roberts, 2013) and complemented this with journals recognized for their scholarly influence and relevance within the field, as reflected in established journal ranking lists such as the 2024 *Academic Journal Guide* (AJG) published by the Chartered Association of Business Schools. The second search included publication titles in the field of Entrepreneurship, Strategy, Organization and General Management journals by using

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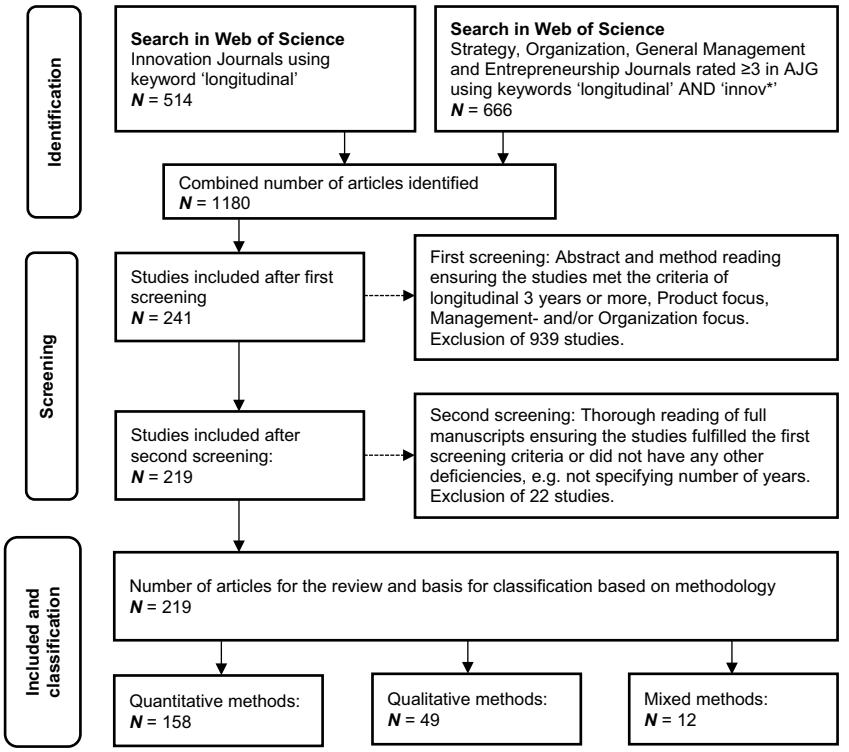


Figure 1. Search and selection strategy

the keyword: "innov*" AND "longitudinal." No lower publication-date limit was imposed; the search therefore covered all available publication years up to May 31, 2025.

After identifying potentially relevant studies, we screened them using three inclusion and exclusion criteria. First, we defined a longitudinal study as one encompassing three years or more, and studies covering shorter periods were excluded. As discussed in Section 2.1, three years was treated as an inclusive minimum threshold rather than an optimal duration for examining all longitudinal innovation processes. Second, because our review is delimited to product innovation, studies focusing exclusively on, for example, process innovation or social innovation unrelated to goods or services were excluded. Third, because the review focuses on resource orchestration within organizations, studies examining innovation primarily at the level of individuals, teams, interorganizational networks, regional innovation systems, or other higher-level units of analysis were excluded. Consequently, the review focuses on resource orchestration within organizations rather than on contextual differences across industries, countries, or innovation ecosystems.

The screening was conducted in two stages. We first examined the abstracts and methodology sections of the identified articles against the eligibility criteria. The articles retained at this stage were subsequently assessed through full-text reading. Both authors jointly and in detail reviewed the first 30 articles, discussing the interpretation and application of the inclusion and exclusion criteria and resolving the different types of borderline cases encountered. After this calibration stage had produced a shared and

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consistently applicable interpretation of the criteria, the remaining articles were randomly divided between the two authors. Any uncertain cases arising during the subsequent screening were discussed jointly until consensus was reached. The procedure resulted in the final inclusion of 219 studies.

3.2 Data analysis

Because the review concerns longitudinal research, we first recorded the number of years covered by each study. Studies based on in-depth analyses of a limited number of cases and studies applying statistical analyses of larger samples generally provide different types of evidence and theoretical insights (Tranfield *et al.*, 2003). In line with previous reviews (Khan *et al.*, 2023), we therefore classified the studies according to their methodological approach. The classification resulted in 49 studies using qualitative methods, 158 studies using quantitative methods and 12 studies using combinations of qualitative and quantitative methods. Subsequently, all studies were mapped onto the resource orchestration framework (Sirmon *et al.*, 2007), which served as the analytical lens for synthesizing the identified longitudinal product innovation studies rather than as an inclusion criterion. Consequently, studies based on different research designs and theoretical perspectives could be synthesized using a common analytical framework while preserving their methodological and epistemological differences. A complete list of the 219 articles included in the review is available from the authors upon request.

For the qualitative studies, we conducted an in-depth reading of the identified articles. The coding was based on the definitions of the orchestration processes presented in Section 2.2. For example, studies focusing on the development of individual resources, such as employee training, were classified as resource accumulation, whereas studies examining organizational learning or other activities involving the combination of several resources were classified as bundling. Studies addressing several orchestration processes were coded in all applicable categories rather than being assigned to a single dominant process.

Concerning the quantitative studies, most used regression-based analyses, and we therefore identified the principal independent and dependent variables in each study. These variables were subsequently mapped onto the resource orchestration framework using the same coding principles as for the qualitative studies. Highly aggregated measures, such as patents or overall innovation performance, were generally interpreted as outcomes of resource orchestration unless they could be linked to a specific orchestration process. This enabled us to determine which orchestration processes, if any, had been examined.

The coding procedure was calibrated through the joint analysis of the first 30 studies by both authors. During this stage, we discussed the interpretation of the orchestration categories and resolved borderline cases until a shared coding framework had been established. The remaining studies were then randomly divided between the authors, and all uncertain classifications were subsequently discussed until consensus was reached. Although the lack of independent coding by both authors represents a limitation of the review, consistency was supported by the initial joint calibration, with any uncertain classifications subsequently discussed jointly until consensus was reached.

4. Results

Qualitative studies mainly used different types of case studies and/or interviews, whereas quantitative studies mainly used secondary data and, to a lesser extent, repeated surveys. Of the twelve studies using combinations of quantitative and qualitative methods, two studies (Cai and Canales, 2024; Huang *et al.*, 2023) used fsQCA. Most other studies using mixed methods used a quantitative analysis and complemented the main results with illustrative

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cases (Willemstein *et al.*, 2007) or interviews with industry experts (Pun *et al.*, 2004). Table 1 illustrates key publication outlets of product innovation management studies with longitudinal research designs.

Most studies using quantitative methods covered time periods between six and ten years and more than half of the studies covered periods shorter than 15 years. Qualitative studies covered shorter periods, with half covering periods shorter than eight years and more than one third examining periods of three or four years. Although studies covering three to five years were included according to the review protocol, considerably longer observation periods were common among quantitative studies. This suggests that longitudinal product innovation research encompasses substantial variation in temporal depth, providing opportunities for future studies to examine how different resource orchestration processes unfold over different time horizons.

4.1 Quantitative studies and resource orchestration

The possibility of considering specific resource orchestration processes is dependent on the type of data that is being used. Figure 2 provides a summary of the data sources used in the 158 quantitative studies.

Table 1. Publication outlets

Journal	Quantitative	Qualitative	Mixed	Total
Research Policy	24	4		28
Technovation	10	13	2	25
Journal of Product Innovation Management	16	5	3	24
Technological Forecasting and Social Change	9	4		13
Journal of Business Research	8	3	1	12
R & D Management	7	4	1	12
Organization Science	9		1	10
International Journal of Innovation Management	5	2		7
Long Range Planning	5	2		7
Academy of Management Journal	7			7
Administrative Science Quarterly	6		1	7
Industry and Innovation	3	3	1	7
Strategic Management Journal	6	1		7
Small Business Economics	6			6
The Journal of Technology Transfer	6			6
Journal of Small Business Management	4	1	1	6
Journal of Business Venturing	4	1		5
Journal of Management Studies	4	1		5
British Journal of Management	3		1	4
International Small Business Journal	2	1		3
Journal of Business Ethics	3			3
Entrepreneurship Theory and Practice	2	1		3
Journal of Management	3			3
European Management Review	2			2
International Journal of Innovation Science	1			1
Family Business Review	1			1
International Journal of Entrepreneurial Behavior & Research	1			1
Organization Studies		1		1
Global Strategy Journal		1		1
Strategic Organization	1			1
Academy of Management Discoveries		1		1
Total	158	49	12	219

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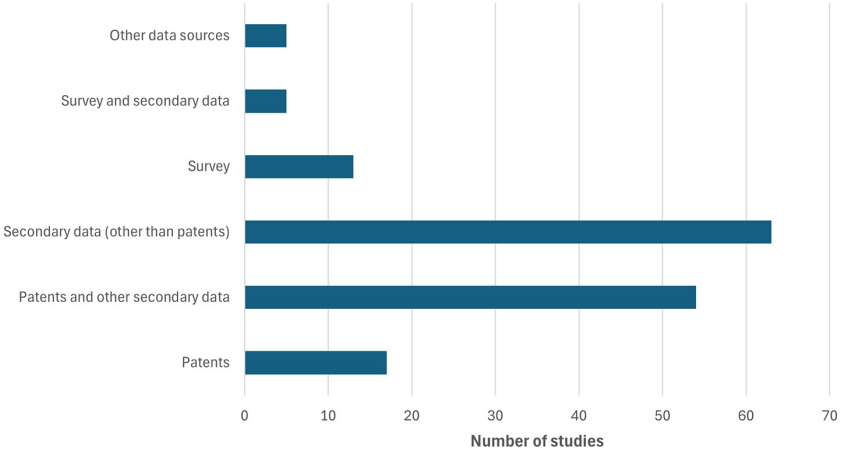


Figure 2. Data sources for quantitative studies

Secondary data refers to data not specifically collected for the study and includes, for example, data from annual reports, broader surveys not collected by the authors and information from other databases such as COMPUSTAT (Harrigan and DiGuardo, 2017), Orbit Intelligence (Ponta *et al.*, 2022), CSMAR (China Stock Market Accounting Research) (Liu *et al.*, 2021) and WIND (Liu *et al.*, 2023). Other data sources include study-specific developed databases based on, for example, video games (De Vaan, 2014) and press releases (Menter *et al.*, 2023). In Figure 2, patents are presented as a separate category because they constitute a distinct and frequently used source of innovation measures in longitudinal product innovation research. Although patent data are archival secondary data, they are separated from other secondary sources to illustrate their particularly prominent role within the reviewed studies.

Most of the quantitative studies examined the overall result of resource orchestration, in terms of innovation outcomes, instead of addressing specific resource orchestration processes. The predominance of quantitative longitudinal studies demonstrates that the field has developed a strong tradition of examining long-term innovation outcomes using established measures and large data sets. Such studies provide valuable evidence regarding the long-term consequences of innovation activities and have contributed substantially to cumulative knowledge on innovation performance. Different types of patent data, such as number of patent approvals (Yang *et al.*, 2023), forward citations of patents (Brem *et al.*, 2016) and collaborative patents (Mueller *et al.*, 2020), are indeed useful measures of innovation and, as evident by our review, patents are an established measure in longitudinal innovation studies. However, because patent measures primarily capture the outcomes of innovation activities rather than the underlying resource management processes, they are generally better suited for examining the consequences of resource orchestration than distinguishing between specific orchestration processes. In addition, most studies combining patent data and secondary data from other sources examined the relationship between patent measures and various overall performance outcomes, such as firm valuation (Zheng *et al.*, 2010), Tobin's Q (Leyva-de la Hiz *et al.*, 2019) and return on sales (Harrigan and DiGuardo, 2017). The use of highly aggregated measures was also evident in studies using secondary data. For example, R&D spending (Wang and Zhou, 2022; Mithani, 2017; Tsai and Wang, 2004), or similar R&D

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related concepts, was used as independent variables in 25 studies. Other studies used other aggregated proxies such as the outsourcing of R&D (Muñoz-Bullón *et al.*, 2020), fraudulent behavior (Wang *et al.*, 2021), ISO-certification (Benner and Tushman, 2002), industry conditions (Visnjic *et al.*, 2019) and firm size (Majumdar, 2011).

Highly aggregated measures are particularly valuable for examining long-term relationships between innovation-related variables across large samples and have contributed substantially to cumulative knowledge on product innovation. However, because such measures primarily capture the outcomes of innovation activities rather than the underlying resource management processes, they rarely distinguish between specific resource orchestration processes such as structuring, bundling and leveraging. Nevertheless, we could identify a limited number of quantitative studies of relevance for both resource structuring, in terms of acquisition (Tsai and Wang, 2008), accumulation (Khurana, 2021) as well as divestment (Grinza and Quatraro, 2019) and the bundling of resources into capabilities (Wang and Chen, 2018). We will address the most relevant studies in more detail in the research agenda section.

4.2 Qualitative studies and resource orchestration

A consolidated overview of the number of qualitative studies addressing each resource orchestration process and subprocess is provided in Appendix Table A1. Of the three main resource orchestration processes, the structuring process has received less attention, being considered in 17 qualitative studies, predominantly by focusing on the subprocess of acquisition. Most of these studies focused on the acquisition of technological resources, such as the innovation implications of acquiring big data (Browder *et al.*, 2022), VoIP (Voice over Internet Protocol) (Cecere, 2009), or secondary innovations (Wu *et al.*, 2009). Only one study (Peltokorpi *et al.*, 2007) considered accumulation of resources in terms of the development of individual resources. Despite that "it is imperative" that firms "actively evaluate current resources and divest less-valued resources to generate the slack and flexibility needed to acquire or accumulate resources of higher value" (Sirmon *et al.*, 2007: 280), none of the qualitative studies examined the divestment of resources. Thus, within the longitudinal product innovation literature reviewed, the structuring process, especially regarding individuals' knowledge, skills and abilities (KSAs), has received modest attention.

The bundling of resources into capabilities was addressed in almost all qualitative studies, and the specific subprocesses of stabilizing/enrichment and pioneering were considered in several studies. Thus, resource bundling in terms of, for example, how the management (Mirvis *et al.*, 2016), creation (Peltokorpi *et al.*, 2007) and integration (Kamuriwo and Baden-Fuller, 2016) of knowledge or inter- (Chiffoleau, 2005; Browder *et al.*, 2022) and intraorganizational (Mirvis *et al.*, 2016; Wu *et al.*, 2009) learning is related to product innovation is a relatively established research field in longitudinal innovation studies. Noticeably, half of the qualitative studies examined both bundling and leveraging.

Like bundling, almost all qualitative studies concerned leveraging and most of these studies addressed mobilization. This subprocess is essentially about identifying and evaluating which capabilities to leverage based on the perceived opportunities in the product market (Sirmon *et al.*, 2007). Because mobilizing capabilities can be initialized based on existing resources, existing market opportunities, or the development of new resources and/or opportunities by entrepreneurial strategies, it makes sense that almost all qualitative studies in our review consider some type of mobilization. Thus, the overall rationale and/or motivation to innovate was considered in almost all studies. However, the specific orchestration subprocess of *coordinating* capabilities with nonstrategic complementary resources and capabilities (Andersén, 2021b; Barney, 1995) was only examined in two

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studies (Sabatini and O'Toole, 2024; Pucci *et al.*, 2020), whereas the final stage of the orchestration process in terms of the physical deployment was specifically considered in seven studies.

5. A resource-orchestration-based research agenda for longitudinal innovation studies

Table 2 presents the proposed future research framework for longitudinal product innovation management by identifying key research gaps emerging from the reviewed literature and developing these, where appropriate, through resource orchestration-based conceptual reasoning into illustrative and actionable research questions. It also suggests complementary theories and methods that may help address these gaps and highlights examples of previous longitudinal studies that can serve as foundations or inspiration for future research. Accordingly, the synthesis focuses primarily on patterns in which resource orchestration processes have received attention, rather than on comparing the substantive findings of studies examining different theoretical relationships and outcomes.

5.1 Structuring

Although structuring constitutes the foundation for resource orchestration, it is the process that has received the least attention in qualitative longitudinal product innovation studies. Most quantitative studies on structuring addressed the structure in terms of, for example, the characteristics of various resources such as the management team (Muñoz-Bullón *et al.*, 2020; Audretsch *et al.*, 2023; Narayan *et al.*, 2021) instead of focusing on the processes of structuring. Building and reconfiguring an organization's resource configuration are time-consuming processes, and generating product innovations through resource structuring is likewise a long-term endeavor. Moreover, capability development is unlikely to follow a linear trajectory, suggesting that future longitudinal studies should examine how resource configurations evolve through periods of gradual development, interruption and renewal. We therefore regard structuring as one of the most important areas for future longitudinal innovation research.

A research area that has received at least some interest is the acquisition of technological resources. Cecere (2009) and Browder *et al.* (2022) provided illustrative longitudinal qualitative studies of technology acquisition. Whereas Cecere (2009) primarily used archival data in terms of industry journals and newspapers, Browder *et al.* (2022) used a wider range of data including interviews and observations. This highlights a more general challenge when conducting longitudinal studies and how studies covering very long periods are often restricted to using secondary data. Whereas qualitative studies have focused on how technology acquisitions can be integrated with other resources, quantitative studies have examined specific antecedents to (Kotlar *et al.*, 2013; Ardito *et al.*, 2018), or aggregated firm-level outcomes (Tsai and Wang, 2008) of, technology acquisition. Examining how technology acquisition is related to other resource orchestration processes over time is therefore a somewhat unexplored but highly relevant research avenue within longitudinal product innovation research. Although few longitudinal studies have examined technology acquisition, future studies should build more explicitly on previous longitudinal research. To contribute to the accumulation of knowledge about long-term processes, as well as consequences, of technology acquisition, it is therefore essential to relate future studies to other quantitative as well as qualitative longitudinal studies on product innovation.

In contrast to technological resources, the structuring of human resources and their KSAs constitutes a major research gap in the longitudinal product innovation literature reviewed that must be addressed to understand long-term resource orchestration for innovation.

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Table 2. A future research framework for longitudinal product innovation management

Type of resource orchestration	Research gap	Example research question	Potential complementary theory or method	Example studies
Structuring	The role of technological resource *acquisition* in longitudinal product innovation	How are technological acquisitions integrated with subsequent resource structuring, bundling and leveraging processes over time?	Strategic factor market theory	Qualitative: Cecere (2009); Browder *et al.* (2022) Quantitative: Ardito *et al.* (2018); Tsai and Wang (2008)
	The role of human resource *structuring* in longitudinal product innovation	How do recruitment, training and retrenchment practices of human resources influence subsequent capability development and product innovation over time?	Strategic human resource management	Qualitative: Peltokorpi *et al.* (2007) Quantitative: Grinza and Quatraro (2019); Sheehan *et al.* (2023)
Bundlin	The relationship between human resource *structuring* and capability *bundling* for product innovation	How does the development of individual KSAs contribute to the emergence of organizational innovation capabilities over time?	Strategic human capital theory	Qualitative: Kamuriwo and Baden-Fuller (2016); Peltokorpi *et al.* (2007) Quantitative: Wang and Chen (2018)
Leveraging	The interplay between capability *bundling* and *leveraging* in longitudinal product innovation	How do capability bundling and capability leveraging evolve over time while organizations balance exploration and exploitation?	Exploration-exploitation ambidexterity	Qualitative: Dixon *et al.* (2014) Quantitative: Narayan *et al.* (2021); Voss and Voss (2013) Mixed methods: Cai and Canales (2024)
	The *coordination* of complementary capabilities in longitudinal product innovation	How are complementary internal and external capabilities coordinated over extended innovation processes?	Absorptive capacity	Qualitative: Pucci *et al.* (2020); Sabatini and O'Toole (2024) Quantitative: Narayan *et al.* (2021);
Overall resource orchestration	The temporal interplay between resource structuring, capability bundling and capability leveraging throughout longitudinal product innovation processes	How do resource structuring, capability bundling and capability leveraging interact and reinforce one another throughout longitudinal product innovation processes?	Ethnography	Qualitative: Fredberg and Piller (2011), Gurdon and Samsom (2010), Wu *et al.* (2009)
	The relative importance of specific resource orchestration processes for different product innovation outcomes	Which resource orchestration processes are most critical for different types of product innovation outcomes?	Resource-based case studies	Qualitative: Capaldo (2007); Sabatini and O'Toole (2024); Van Geenhuizen and Soetanto (2009)

Although the structuring of human resources was only addressed in one qualitative study (Peltokorpi *et al.*, 2007), there are a few quantitative studies addressing various structuring subprocesses related to human resources. Concerning the acquisition of human resources, quantitative longitudinal studies have found that hiring scientists in biotechnology firms attracts more R&D collaborations (Luo *et al.*, 2009) and have highlighted the importance of the skillset of employees for high-tech companies (Siepel *et al.*, 2017). Moreover, accumulation in terms of training has been found to be beneficial for both incremental and radical innovations (Sheehan *et al.*, 2023). When it comes to divestment of human resources, the notion of Sirmon *et al.* (2007: 280–281) that because top management "are often unaware of resources' future value… extreme caution should be exercised in divesting resources" also holds true for innovation. Specifically, numerical flexibility of the workforce was found to have a negative impact on both incremental and radical innovations (Sheehan *et al.*, 2023), and Grinza and Quatraro (2019: 1) found that "workers' replacements significantly dampen innovation performance." Although human resource management has received some interest in product innovation management research (Beugelsdijk, 2008; Haneda and Ito, 2018; Krammer, 2022; Seeck and Diehl, 2017) and the nature of the long-term processes associated with bundling human resources into organizational capabilities and leveraging those capabilities to generate product innovation, human resource management has rarely been applied in longitudinal innovation studies. Thus, future longitudinal research should build on HRM and innovation research to examine recruitment, skill development and retrenchment management and their long-term effects on product innovation.

5.2 Bundling

Exploration of new knowledge is a highly established area in innovation management research and most qualitative studies and several quantitative studies (Wang and Chen, 2018; George *et al.*, 2008; Katila and Ahuja, 2002) in the review considered the bundling of resources into capabilities. Therefore, the resource orchestration framework offers limited guidance on how to advance this specific element of orchestration. Instead, an important future research avenue for resource bundling is to undertake studies focusing on the relationship between the structuring and bundling of internal and external resources. Absorptive capacity could provide a particularly useful complementary perspective here, as it explicitly addresses the acquisition and utilization of external knowledge, whereas resource orchestration does not distinguish systematically between the management of internally and externally developed resources.

Specifically, more knowledge is needed on how the development of individuals' KSAs contributes to bundling resources into organizational-level capabilities. We identified several longitudinal studies (Grinza and Quatraro, 2019; Sheehan *et al.*, 2023) showing that individual resources are important for product innovation, but not how they are transformed into organizational capabilities. Moreover, exploration of organizational knowledge ultimately depends on individuals' KSAs. Peltokorpi *et al.* (2007) remains the only qualitative longitudinal study addressing this research area. By collecting data from a period of six years, this study adapted the knowledge creation model developed by Nonaka (1994), on how individual knowledge transforms into team and organizational knowledge, to innovation processes. Strategic human capital theory could also provide a useful complementary perspective. Ployhart *et al.* (2014) distinguished between individual and organizational human capital, and the emergence of organizational-level human capital remains a central research area (Ray *et al.*, 2023). Thus, drawing on strategic human capital theory could facilitate research on the interplay between structuring and bundling of human resources.

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5.3 Leveraging

Like bundling, leveraging was considered in most qualitative and several quantitative studies, and many examined both processes. Thus, overall leveraging can be regarded as an established research area in longitudinal product innovation management studies. Recurring perspectives include ambidexterity (Choi *et al.*, 2022; Voss and Voss, 2013), dynamic capabilities (Dixon *et al.*, 2014; Sheehan *et al.*, 2023) and absorptive capacity (Zahra and George, 2002). Most studies build primarily on earlier work using the same theoretical perspectives. However, given the limited number of longitudinal innovation studies, future studies should build more explicitly on previous longitudinal research to advance cumulative knowledge of long-term innovation processes.

Although leveraging is well established, the coordination of capabilities with complementary capabilities has received limited attention in qualitative longitudinal studies. Complementary capabilities enable organizations to fully leverage valuable capabilities (Barney, 1995; Barney, 1997) to generate product innovations. Whereas several quantitative studies addressed coordination by considering various moderating and mediating factors (Kyriakopoulos, 2011; Cheng and Shiu, 2022; Liu *et al.*, 2023), only two qualitative studies (Sabatini and O'Toole, 2024; Pucci *et al.*, 2020) examined coordination. Both were single-case studies based primarily on archival data and retrospective interviews, focusing on the coordination of external capabilities. These studies suggest that the coordination of complementary capabilities is a complex, long-term process and therefore represents a promising area for future innovation research.

5.4 Overall resource orchestration

Although the proposed future research framework identifies research gaps related to specific resource orchestration processes, our review also revealed limited integration of orchestration processes over time. Resource orchestration is ultimately concerned with the integration and orchestration of all processes. Because resource orchestration encompasses multiple interrelated processes, more holistic approaches are most likely to benefit from at least some elements of qualitative-oriented research designs. We can see two promising research areas for more holistic longitudinal studies on product innovation using resource orchestration.

First, longitudinal studies offer an opportunity for in-depth studies that consider all elements of the resource orchestration process. Because product innovation ultimately results in tangible innovation outcomes, longitudinal studies offer unique opportunities to examine how different orchestration processes contribute to these outcomes over time. Specifically, processes toward the end of this realization, such as leveraging, can be examined by analyzing shorter time periods, whereas resource structuring is likely to require particularly long observation periods. Whereas perspectives such as absorptive capacity and dynamic capabilities provide important insights into particular aspects of bundling and leveraging, resource orchestration is particularly useful for longitudinal research concerned with the entire resource management process, including the long-term structuring of the resource portfolio. Indeed, applying this lens made visible the limited attention devoted to structuring in longitudinal product innovation research. Future longitudinal studies should also pay greater attention to how resource orchestration processes are sequenced, overlap and interact over time, rather than assuming linear progressions from structuring to bundling and leveraging. In addition, examining multiple orchestration processes necessary to promote product innovation, and the complexity of cumulative and recursive relationships between them, is likely to require vast amounts of data from respondents in terms of interviews and/or observations and could benefit from using, in innovation management research, less

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common research approaches such as ethnography (Fernandes and Burcharth, 2024; Vesa and Vaara, 2014). Although such research designs are time consuming, they are feasible. For example, Wu *et al.* (2009: 399) conducted a study based on "nearly 20 years of continuous academic observation," including "plant tours and semi-structured/open-ended interviews," and other studies conducted interviews over a seven year period (Fredberg and Piller, 2011; Wikhamn and Styhre, 2023). However, there are alternatives to continuous data collection over longer periods of time. For example, by conducting follow-up studies several years after initial studies (Gurdon and Samsom, 2010), by conducting most of the data collection at one point and complementing with data from previous interviews and archival data (Pucci *et al.*, 2020), or by conducting periodic follow-up studies of previous research projects over extended periods (e.g. every third year). Because the purpose of this review was to synthesize resource orchestration processes rather than compare contextual conditions, contextual characteristics such as industry, country and ecosystem embeddedness were not systematically analyzed. Future longitudinal research could therefore examine how resource orchestration processes differ across organizational and institutional contexts, thereby extending the contextual understanding of longitudinal product innovation management. At the same time, qualitative and quantitative approaches should be regarded as complementary rather than competing. Whereas qualitative studies are particularly valuable for uncovering how resource orchestration processes unfold over time, quantitative longitudinal studies provide robust evidence regarding the antecedents and long-term consequences of these processes across larger samples.

The second research area concerns the identification of specific critical orchestration processes, and/or combinations of specific processes, for different types of product innovation outcomes. For example, radical product innovation is likely to depend on different orchestration processes than incremental innovation, while organizations operating in short product life cycles are likely to emphasize different orchestration processes than firms relying on more stable product portfolios. Resource orchestration can here be applied retrospectively to identify whether it is mainly the efficient leveraging of capabilities that enables an organization to excel in their innovation, or if the innovation outcomes can best be explained by some specific capabilities, and, if so, if these capabilities are the result of some specific and unique resources or the bundling of more generic resources. The research methodology suggested by Rouse and Daellenbach (1999) of identifying companies based on their performance outcomes and then conducting in-depth case studies of these companies can be used to study this issue. The step of identifying firms can, for example, involve "the selection of an industry and generating performance data and rankings from secondary sources" (Rouse and Daellenbach, 1999: 489). Archival data from secondary sources and retrospective interviews can then be undertaken to identify and examine the nature of specific orchestration processes. Rather than relying solely on secondary data, relevant companies can also be identified through more detailed quantitative analyses, for example by using cluster analysis. Thus, mixed methods can be a viable option for addressing this research area (Pun *et al.*, 2004; Willemstein *et al.*, 2007). This research area can also use purely qualitative methods and several qualitative studies used retrospective interviews (Sabatini and O'Toole, 2024; Van Geenhuizen and Soetanto, 2009). The study by Capaldo (2007) is especially interesting because of the care taken to identify relevant case companies. By exploratory interviews with "CEOs and managers of numerous firms, and several industrial designers and industrial design scholars," Capaldo (2007: 591) identified three highly innovative organizations. These were then studied using various data sources, including archival data and retrospective interviews.

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6. Conclusion

This review has addressed longitudinal studies in product innovation management research by analyzing studies of three years or longer in journals in innovation and related fields. Despite a slight increase in the number of studies over the years, longitudinal product innovation studies – especially those using qualitative research designs – remain modest in number, and most have examined shorter time spans. From a managerial perspective, the findings suggest that managers should view product innovation as a long-term resource orchestration process rather than a series of isolated innovation projects. In particular, the review highlights human capital development, capability renewal and the coordination of complementary resources as potentially important areas for maintaining long-term product innovation performance.

By viewing product innovation through a resource orchestration approach, we have identified resource structuring as an under-researched yet highly promising area for future longitudinal product innovation management research. Longitudinal studies offer unique opportunities to examine the long-term processes needed to develop the foundation for firms' capabilities, particularly the management of an organization's resource configuration, especially human resources and their KSAs. More in-depth studies focusing on structuring *per se*, or on the relationships and roles of all resource orchestration processes for product innovation, would considerably advance our understanding of how companies can orchestrate their resources to generate new or renewed products. Our review also suggests that the appropriate temporal horizon for longitudinal product innovation research is likely to vary depending on the particular resource orchestration process under investigation, as structuring, bundling and leveraging are likely to unfold over different time periods. We hope that our review stimulates further discussion and development on conducting longitudinal product innovation studies and inspires future research endeavors.

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Appendix

A1. Search strategy in Web of Science.

Search terms: "longitudinal" (innovation journals); "innov*" AND "longitudinal" (non-innovation journals).

Search fields: Title, Key words, Abstract.

Search period: All available publication years up to May 31, 2025

Innovation journals: Research Policy, *Journal of Product Innovation Management*, Industry and Innovation, *Journal of Technology Transfer*, R&D Management, Technological Forecasting and Social Change, Technovation, *International Journal of Innovation Management*, *International Journal of Innovation Science*.

Other journals: All journals rated by the 2024 *Academic Journal Guide* (AJG) as 3 or 4 in the fields of Entrepreneurship, Strategy, Organization and General Management.

Table A1. Distribution of resource orchestration processes across qualitative studies

Number of studies addressing…	Resource orchestration process	#
… overall or some element of main resource orchestration processes	Structuring	17
	Bundling	40
	Leveraging	45
… single or combinations of main resource orchestration processes	Only structuring	0
	Only bundling	1
	Only leveraging	5
	Structuring and bundling	1
	Structuring and leveraging	2
	Bundling and leveraging	24
	Structuring and bundling & leveraging	14
… structuring sub-processes	Acquiring	12
	Accumulating	1
	Divesting	0
… bundling sub-processes	Stabilizing/Enriching	28
	Pioneering	20
… leveraging sub-processes	Mobilizing	39
	Coordinating	2
	Deploying	7

Note(s): Studies addressing more than one resource orchestration process or sub-process were coded in all applicable categories. Consequently, counts for individual processes and sub-processes are not mutually exclusive

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